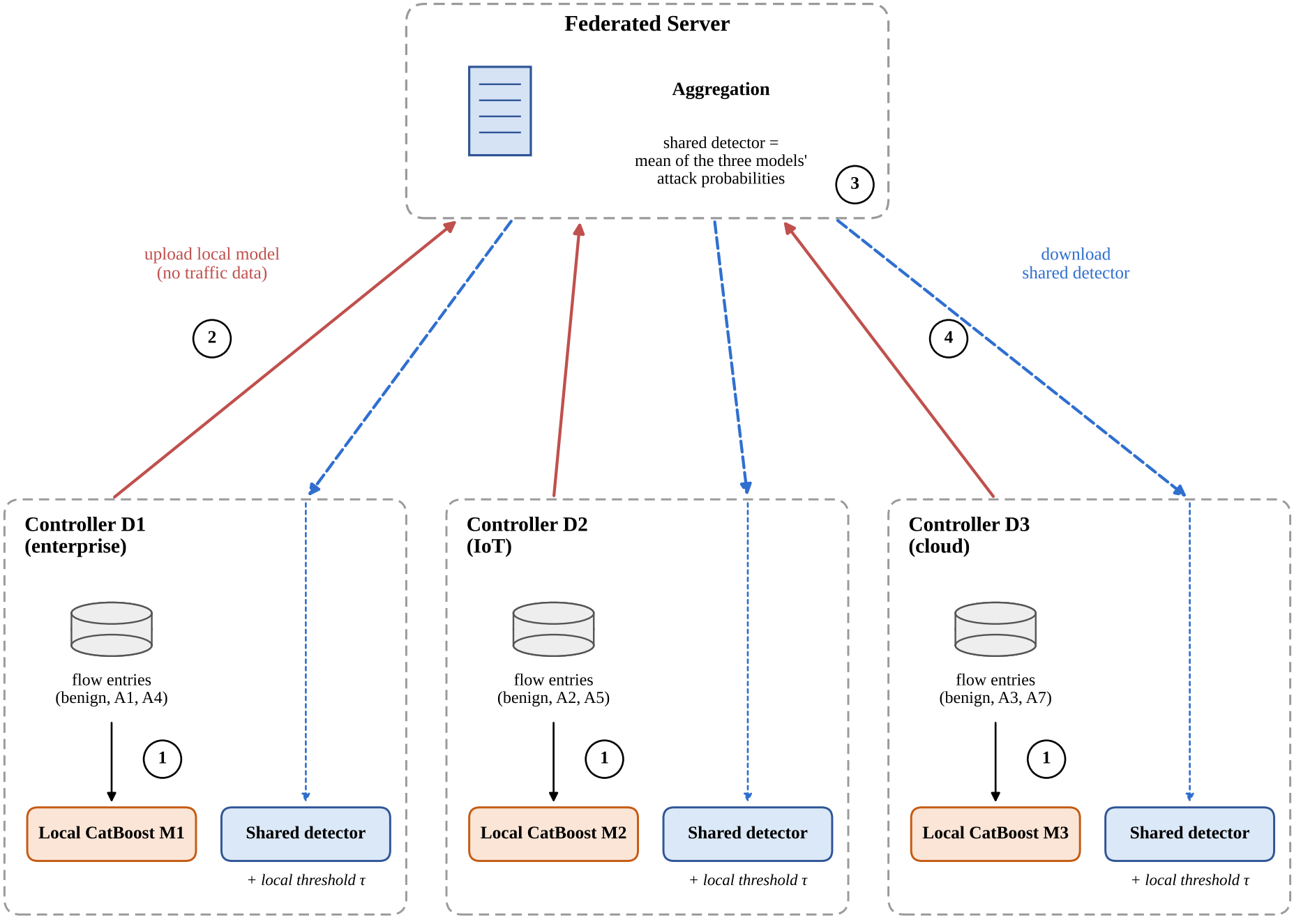
Federated Server
Aggregation
shared detector =
mean of the three models'
attack probabilities
3
upload local model
(no traffic data)
2
download
shared detector
4
Controller D1
(enterprise)
flow entries
(benign, A1, A4)
1
Local CatBoost M1
Shared detector
+ local threshold τ
Controller D2
(IoT)
flow entries
(benign, A2, A5)
1
Local CatBoost M2
Shared detector
+ local threshold τ
Controller D3
(cloud)
flow entries
(benign, A3, A7)
1
Local CatBoost M3
Shared detector
+ local threshold τ